Formula Student AI Endurance Invitational Regulations 2026

1. Scope and Intent

These regulations define the technical and sporting framework for the Formula Student AI Endurance Invitational, a closed-course engineering simulation event for university prototype race cars. The rules are written to resemble a professional motorsport handbook while remaining suitable for student design review, technical scrutineering, and AI-assisted compliance analysis.

All teams are responsible for presenting a vehicle that is safe, dimensionally compliant, and technically inspectable. Where a rule sets a measurable limit, that limit shall be treated as absolute unless an explicit tolerance is stated.

2. Vehicle Safety

2.1 Primary structures

The vehicle shall incorporate a front bulkhead, front impact structure, main roll hoop, front roll hoop, and side-impact structure capable of maintaining survival space around the driver. No aerodynamic or bodywork element may reduce the required driver egress opening.

2.2 Driver egress

The driver shall be capable of exiting the fully assembled vehicle within five seconds without the use of tools. Steering wheel quick-release systems shall be positively locked in the installed condition.

2.3 Shutdown systems

The car shall include an externally accessible emergency shutdown actuator on each side of the chassis and one driver-operated shutdown actuator within the cockpit. Each actuator shall isolate tractive power and disable torque delivery.

2.4 Fire and thermal safety

Any component within 100 mm of the accumulator container, inverter, or high-current bus shall be self-extinguishing or metallic. Exposed edges within driver reach shall have a minimum radius of 2.0 mm.

3. Aerodynamic Bodywork

3.1 General envelope

All aerodynamic devices shall remain within the approved bodywork envelope. No element may project beyond the vehicle width limits established in Section 8. Exposed edges on aerodynamic devices shall have a radius of at least 3.0 mm.

3.2 Rear wing geometry

The complete rear wing assembly, including endplates, shall not exceed 1200 mm total width. The maximum chord of any rear-wing element shall be 300 mm. No aerodynamic surface associated with the rear wing may extend more than 250 mm rearward of the rear axle centerline.

3.3 Endplates and supports

Each rear-wing endplate shall fit entirely inside an 820 mm by 420 mm rectangle when viewed in side elevation. Wing supports shall be structural members only and shall not contain active or movable aerodynamic devices.

3.4 Front wing and canards

No front aerodynamic device may extend more than 700 mm forward of the front axle centerline. Flexible or driver-adjustable front wing systems are prohibited.

3.5 Suspension fairings

Suspension members may use fairings for drag reduction, but any fairing wrapped around a suspension arm shall not exceed 35 mm maximum chord and 20 mm maximum thickness. Fairings shall not create a closed lifting profile around a structural member.

4. Tyres and Wheels

4.1 Approved tyre categories

Only dry slick, intermediate, or wet tyres supplied in the approved competition sizes may be used. Chemically altered tyres and manually cut slicks are prohibited.

4.2 Operating limits

Teams may define target hot pressures, but no tyre may exceed 1.85 bar hot pressure as measured immediately after a run. Tyre blankets and active external tyre-heating devices are prohibited in the event paddock.

4.3 Tyre identification

Each tyre compound introduced in scrutineering shall be uniquely identified and traceable in engineering records. Teams shall maintain a wear model for each approved compound used during dynamic sessions.

5. Powertrain and Energy Storage

5.1 Propulsion system

The vehicle shall use an electric powertrain with a maximum continuous delivered power of 85 kW and a maximum peak delivered power of 110 kW for up to ten seconds. Torque-vectoring logic is permitted if it remains software-limited and fail-safe.

5.2 Accumulator container

The accumulator container shall be mechanically independent from removable bodywork and shall remain sealed during dynamic events. The enclosure shall provide electrical isolation, debris resistance, and containment under foreseeable thermal failure conditions.

5.3 Permitted enclosure materials

Primary accumulator enclosure walls shall be constructed from one of the following:

- Aluminum alloy of the 5xxx or 6xxx series with minimum wall thickness of 2.0 mm.
- Steel with minimum wall thickness of 1.0 mm.
- Composite laminate using self-extinguishing resin with an aramid inner layer and demonstrated burn-through resistance equivalent to the metallic options above.

Magnesium, magnesium-lithium alloys, wood, and unprotected commodity thermoplastics shall not be used as the primary load-bearing enclosure material.

5.4 Electrical isolation

The accumulator container shall maintain insulation resistance and shall incorporate isolation monitoring. Tractive-system components shall be enclosed so that accidental contact cannot occur during service operations.

6. Mass, Ballast, and Minimum Weight

The complete vehicle in running order, excluding the driver, shall weigh not less than 235 kg. Any ballast shall be positively retained with mechanical fasteners and shall not be mounted to aerodynamic bodywork.

7. Materials and Construction

7.1 Structural materials

Load-bearing chassis and suspension components shall be manufactured from metallic materials or validated composite laminates with documented layups. Brittle castings in primary suspension load paths are prohibited unless their design loads are independently certified.

7.2 Fasteners and joints

All suspension joints shall use positive retention. Adhesive bonding may supplement mechanical retention but shall not be the sole retention method for suspension or steering systems.

7.3 Heat-affected areas

Materials exposed to brake disc radiation, inverter cooling hardware, or accumulator ventilation paths shall maintain their rated properties at the maximum expected service temperature.

8. Dimensions and Tolerances

8.1 Vehicle box

The complete car shall fit inside a box measuring 3300 mm long and 1600 mm wide. The maximum overall height is 1250 mm measured from the ground with the car in running condition.

8.2 Wheelbase and track

Wheelbase shall be between 1525 mm and 1725 mm. Front track and rear track shall each be between 1150 mm and 1320 mm.

8.3 Suspension members

Metallic wall thickness of a load-bearing suspension tube or insert shall not be less than 1.0 mm. Composite suspension fairings shall preserve visibility of inspection points and shall be removable without cutting.

8.4 Ground clearance

The vehicle shall maintain a static ground clearance of at least 28 mm with the driver absent and the accumulator at nominal state of charge.

9. Testing and Scrutineering Procedures

Every team shall present a technical dossier containing major dimensions, material callouts, and an itemized list of experimental parts. Any component not described in the dossier may be rejected from use until re-inspection is completed.

Scrutineering officials may request CAD snapshots, laminate schedules, or material certificates for any safety-critical part. If a measured dimension exceeds the stated tolerance, the component shall be treated as non-compliant until corrected.

10. Sporting Procedures and Penalties

10.1 Session control

Race control may use local yellow, full-course yellow, or safety-car procedures. Teams must adapt strategy recommendations to the active session control state and shall not treat pace data from neutralized laps as representative green-flag pace.

10.2 Penalty framework

Technical non-compliance may result in deletion of laps, exclusion from the session, or refusal of permission to run. Sporting violations may result in time penalties, drive-through equivalents in simulation, or forfeiture of classification.

10.3 Final authority

The chief scrutineer and race director retain final authority over safety and sporting interpretation. Where ambiguity exists, the safer interpretation shall prevail.

11. Interpretation Note

This handbook is intentionally structured for retrieval tasks. Compliance answers should cite the relevant section numbers, compare the component specification to the applicable limit, and state whether the part is compliant, non-compliant, or unsupported by the available evidence.